AdaptDuplex and Version 8

How to walk the two flowcharts

AdaptDuplex is a research model from September 2026. Listening and speaking are trained together. Version 8 is our localhost. Recognition, the language model, and the voice are three published Sarvam services. We do not train them.

AdaptDuplex

The Thinker and the Talker are trained together, on Qwen3-Omni. Listening continues during the reply.

1. **User audio**

   Sound keeps arriving in a short window, 0.48, 0.64, or 0.96 seconds, including while the agent is speaking. There is no wait for a full silence.

2. **Thinker, trained**

   This is the language model plus its audio encoder, Qwen3-Omni-30B-A3B. It decides whether to listen or speak, how long the next window is, and the reply text. Training is in three stages: ordinary turns, then pauses and interruptions, then memory and tool tokens.

3. **Talker, trained with the Thinker**

   It reads the Thinker's hidden states and writes speech tokens. The text may lead the voice by at most two tokens. It is trained alone first, then together with the Thinker. An optional reinforcement step can follow.

4. **System audio**

   The waveform plays while the next window is already going back into the Thinker. The side arrow is labeled still listening.

Recognition, the language model, and the voice are one system. The training set is 765,424 synthetic dialogues, 17,259 hours, then a smaller joint fine-tune on 8 H100 GPUs. On human-recorded HumDial-FDBench the Final score is 69.6, the highest among the duplex models that paper compares.

Version 8

Start at the title: hear, then decide, then speak. One clip is recognised, then the reply is spoken.

1. **User audio**

   A turn starts after about 0.4 seconds of speech and is sent after about one second of quiet. The microphone is 16 kHz mono.

2. **Speech recognition, used as published**

   Sarvam Saaras v4 transcribes that one clip. We do not train it.

3. **Language model, used as published**

   A fixed line, such as the sales-advisor close, is spoken without a model call. A free reply is one non-streaming call to sarvam-105b-conversations. Written rules choose the stage, the facts, and the close.

4. **Voice, used as published**

Sarvam Bulbul v3, speaker Ishita. The first clause starts playing while the later clauses are prepared. Pace is speed and temperature is warmth. Neerja speaks only when Bulbul returns no audio. “sq. ft.” is spoken as “square feet” and “SOBHA” as “Sobha.” The written reply stays unchanged. Optional - A goodbye is a little slower, 0.9 and 0.7. A visit is a little quicker, 1.05 and 0.65. A size or a price is steadier, 0.95 and 0.55.

The side arrow is labeled barge-in. Playback pauses after about 0.4 seconds of louder speech. The reply is not being generated and listened to as one loop.

Under the figure. The three services stay separate. This project does not train them. A fixed line never calls the model. Ishita speaks the first clause while the rest of the reply is prepared.

The difference

AdaptDuplex keeps a window of the caller’s voice going into the Thinker while its own voice is playing, and that behaviour was trained. Version 8 finishes the caller’s turn, then recognises, then answers, then speaks. A loud interruption can pause Ishita, and the next turn starts again from the top.